Chapter 7, Problem 85P

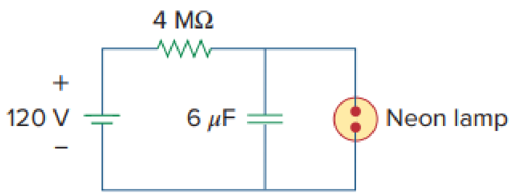
Problem

A simple relaxation oscillator circuit is shown in Fig. 7.145. The neon lamp fires when its voltage reaches 75 V and turns off when its voltage drops to 30 V. Its resistance is 120 Ω when on and infinitely high when off.

(a) For how long is the lamp on each time the capacitor discharges?

(b) What is the time interval between light flashes?

Fig. 7.145:



Step-by-step solution

Step 1 of 3

(A) The light is on from 75 volts until 30 volts during that time we essentially have a 120 Ω resistor in parallel with a 6 μ F capacitor.

V(0) = 75

V(∞) = 0

$$\tau = 120 \times 6 \times 10^{-6} = 0.72\,mS\ .$$

$$V(t_1) = 75\,e^{\frac{-t_1}{\tau}} = 30$$

Which leads to,

t1 = -0.72 ln(0.4) mS.

= 659.7 μS. Of lamp time.

Step 2 of 3

(B) $\tau = RC = \left(4\times10^{6}\right)\left(6\times10^{-6}\right) = 24\,S$ Since,

$$V(t) = V(\infty) + [V(0)-V(\infty)]\,e^{\frac{-t}{\tau}}$$

$$V(t_1) - V(\infty) = [V(0)-V(\infty)]\,e^{\frac{-t_1}{\tau}} \rightarrow (1)$$

$$V(t_2) - V(\infty) = [V(0)-V(\infty)]\,e^{\frac{-t_2}{\tau}} \rightarrow (2)$$

Step 3 of 3

Divide (1) by (2),

$$\frac{V(t_1)-V(\infty)}{V(t_2)-V(\infty)} = e^{\frac{(t_2-t_1)}{\tau}}$$

$$t_0 = t_2 - t_1 = \tau \ln\left[\frac{V(t_1)-V(\infty)}{V(t_2)-V(\infty)}\right]$$

$$t_0 = 24 \ln\left[\frac{75-120}{30-120}\right] = 24 l\text{n}(2) = 16.636S$$